

Revenue Debt Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Revenue Debt Discipline (RDD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on RDD achieves an annualized gross (net) Sharpe ratio of 0.46 (0.34), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 18 (13) bps/month with a t-statistic of 2.62 (2.01), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, Employment growth) is 15 bps/month with a t-statistic of 2.24.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not immediately reflect all available information, challenging the traditional efficient market hypothesis. While classical finance theory assumes instantaneous price adjustment to new information, behavioral and institutional frictions create systematic delays in how markets incorporate fundamental signals into asset prices. We identify a novel predictor of cross-sectional stock returns based on Revenue Debt Discipline (RDD), which captures firms' ability to manage debt obligations relative to their revenue generation capacity. This signal exploits the gradual diffusion of information about corporate financial discipline, particularly how efficiently firms convert revenue streams into debt management capabilities. Despite the fundamental importance of debt-revenue relationships for firm valuation, investors appear to systematically underreact to this information, creating predictable patterns in future returns.

The slow diffusion of information provides a compelling economic mechanism linking Revenue Debt Discipline to future stock returns. Following seminal work on limited attention and gradual information incorporation [Hong and Stein \(1999\)](#), [DellaVigna and Pollet \(2009\)](#), and [Hirshleifer et al. \(2003\)](#), we hypothesize that investors systematically underreact to signals embedded in the relationship between revenue generation and debt management. When firms demonstrate superior revenue-to-debt discipline, this information diffuses gradually through the market as different investor segments process the signal at varying speeds. [Hong et al. \(2000\)](#) document that bad news travels slowly, particularly for stocks with low analyst coverage, while [Peng and Xiong \(2006\)](#) show that investors face capacity constraints that force them to allocate attention sequentially rather than simultaneously across different information sources. The complexity of interpreting debt-revenue relationships exacerbates these attention constraints, as investors must synthesize information from

multiple financial statement items to assess the signal’s implications.

The gradual incorporation of RDD information should manifest most strongly where informational frictions are severe. [Hou and Moskowitz \(2005\)](#) demonstrate that market frictions delay price discovery, particularly among smaller, less liquid stocks where transaction costs and limits to arbitrage prevent rapid price adjustment. Building on [Cohen and Frazzini \(2008\)](#)’s evidence of predictable underreaction to economically linked information, we expect RDD’s predictive power to concentrate among firms with limited investor attention, measured by factors such as size, analyst coverage, and institutional ownership. The signal should predict return continuation rather than reversal, consistent with initial underreaction followed by gradual price adjustment as information disseminates across the investor base.

Furthermore, the multifaceted nature of Revenue Debt Discipline creates additional barriers to rapid information processing. Unlike simple accounting ratios, RDD requires investors to jointly evaluate revenue trends and debt dynamics, creating what [Cohen et al. \(2012\)](#) characterize as complicated information that investors struggle to process efficiently. [Bernard and Thomas \(1989\)](#) document post-earnings-announcement drift arising from investors’ failure to fully incorporate earnings information, and we expect similar underreaction to RDD signals that combine revenue and debt metrics. As [Tetlock et al. \(2008\)](#) show, investor inattention to public information generates return predictability, particularly when the information requires sophisticated analysis to extract its valuation implications. The RDD signal exploits precisely this type of complex, multi-dimensional information that investors process slowly and incompletely.

We find strong evidence that Revenue Debt Discipline predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on RDD achieving an annualized gross Sharpe ratio of 0.46 and net Sharpe ratio of 0.34 after accounting for transaction costs. The strategy generates a monthly average gross

return of 23 basis points with a t-statistic of 3.27, demonstrating robust statistical significance. Most notably, the RDD strategy maintains significant abnormal returns after controlling for established risk factors, with monthly alphas ranging from 18 to 29 basis points across standard factor models. The alpha relative to the [Fama and French \(2018\)](#) six-factor model equals 18 basis points per month with a t-statistic of 2.62, indicating that RDD captures a distinct source of return predictability not explained by market, size, value, profitability, investment, or momentum factors.

The economic significance of RDD remains substantial after accounting for implementation costs and alternative portfolio construction methods. Net returns after transaction costs yield 17 basis points per month with a t-statistic of 2.46 for the baseline value-weighted NYSE-breakpoint strategy, while the net generalized alpha relative to the six-factor model equals 13 basis points with a t-statistic of 2.01. Critically, the predictive power of RDD concentrates among larger, more liquid stocks where implementation is feasible—among stocks above the 80th NYSE size percentile, the strategy generates 29 basis points per month with a t-statistic of 3.18. This concentration of returns among tradeable stocks distinguishes RDD from many documented anomalies that exist primarily among micro-cap securities.

RDD demonstrates remarkable robustness to controls for related anomalies and maintains its predictive power within the broader factor zoo. When we control for the six most closely related strategies—Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, and Employment growth—the RDD strategy retains a significant alpha of 15 basis points per month with a t-statistic of 2.24. The strategy’s gross Sharpe ratio of 0.46 exceeds 86% of anomalies in the factor zoo, while its net Sharpe ratio of 0.34 surpasses 93% of existing strategies after transaction costs. Furthermore, adding RDD to a combination strategy incorporating 155 existing anomalies increases terminal wealth from 993.77to

1,138.20, demonstrating that RDD provides incremental value beyond the collective information in documented return predictors.

This paper makes several contributions to the asset pricing literature on information diffusion and cross-sectional return predictability. We extend the gradual information diffusion framework of [Hong and Stein \(1999\)](#) by identifying a novel signal that captures the complex interaction between revenue generation and debt management, demonstrating that investors systematically underreact to this multi-dimensional information. Unlike [Sloan \(1996\)](#)’s accrual anomaly or [Cooper et al. \(2008\)](#)’s asset growth effect, which focus on single accounting dimensions, RDD synthesizes information across financial statement categories to measure firms’ financial discipline. Our findings complement [Novy-Marx \(2013\)](#)’s gross profitability premium and [Fama and French \(2015\)](#)’s quality factors by showing that the relationship between revenues and debt obligations contains distinct information about future returns not captured by existing profitability or investment measures.

Methodologically, we employ the rigorous testing framework of [Novy-Marx and Velikov \(2023b\)](#) to establish RDD’s robustness across multiple dimensions, including alternative portfolio constructions, transaction cost implementations, and controls for the extensive factor zoo. This comprehensive approach addresses concerns raised by [Harvey et al. \(2016\)](#) about data mining and [McLean and Pontiff \(2016\)](#) about publication bias in anomaly research. By demonstrating that RDD maintains significant predictive power after controlling for 210 existing anomalies and generates positive net returns after transaction costs, we provide evidence that the signal represents a genuine source of mispricing rather than a statistical artifact. The concentration of returns among larger, liquid stocks further distinguishes our findings from many anomalies that [Hou et al. \(2020\)](#) show exist primarily among difficult-to-arbitrage securities.

Our results have important implications for understanding market efficiency and

the limits of arbitrage in modern financial markets. The persistence of RDD’s predictive power, particularly among larger stocks with substantial institutional ownership, suggests that even sophisticated investors fail to fully process complex, multidimensional signals about firm fundamentals. This evidence supports theoretical models of limited attention such as [Sims \(2003\)](#) and [Peng and Xiong \(2006\)](#), which predict that capacity constraints force investors to process information sequentially rather than simultaneously. The economic magnitude of the RDD premium—generating net alphas exceeding 13 basis points per month after costs and factor adjustments—indicates that informational inefficiencies create meaningful opportunities for investors who can identify and exploit gradual information diffusion. These findings underscore the importance of understanding how market participants process complex financial information and highlight the continued relevance of fundamental analysis in an era of increasing market sophistication.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative sales values to ensure meaningful scaling. Revenue Debt Discipline signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that increase a firm’s long-term obligations. This item reflects the gross amount of long-term debt issued rather than net changes in debt levels. Sales (SALE) represents the total revenue generated from a firm’s primary business operations during the fiscal year, providing

a measure of firm size and operating scale that serves as an appropriate denominator for scaling financial activities. We construct the Revenue Debt Discipline signal by calculating the year-over-year change in long-term debt issuance scaled by lagged sales: $(DLTIS(t) - DLTIS(t-1)) / SALE(t-1)$. This measure captures the change in a firm’s debt issuance activity relative to its revenue-generating capacity, with positive values indicating acceleration in debt issuance and negative values suggesting deceleration or increased debt discipline. The scaling by lagged sales ensures comparability across firms of different sizes and normalizes the debt issuance changes by the firm’s economic scale. We compute this signal using annual observations at each fiscal year-end and require non-missing values for both DLTIS and SALE in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RDD signal. Panel A plots the time-series of the mean, median, and interquartile range for RDD. On average, the cross-sectional mean (median) RDD is -0.47 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input RDD data. The signal’s interquartile range spans -0.12 to 0.15. Panel B of Figure 1 plots the time-series of the coverage of the RDD signal for the CRSP universe. On average, the RDD signal is available for 6.12% of CRSP names, which on average make up 7.43% of total market capitalization.

4 Does RDD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RDD using NYSE breaks. The first two lines of Panel A report

monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RDD portfolio and sells the low RDD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short RDD strategy earns an average return of 0.23% per month with a t-statistic of 3.27. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.18% to 0.29% per month and have t-statistics exceeding 2.62 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.35, with a t-statistic of 7.53 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 526 stocks and an average market capitalization of at least \$1,687 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns

to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 16 bps/month with a t-statistics of 3.39. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -10-24bps/month. The lowest return, (-10 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RDD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the RDD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RDD, as well as average returns and alphas for long/short trading RDD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the RDD strategy achieves an average return of 29

bps/month with a t-statistic of 3.18. Among these large cap stocks, the alphas for the RDD strategy relative to the five most common factor models range from 18 to 34 bps/month with t-statistics between 1.99 and 3.69.

5 How does RDD perform relative to the zoo?

Figure 2 puts the performance of RDD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the RDD strategy falls in the distribution. The RDD strategy’s gross (net) Sharpe ratio of 0.46 (0.34) is greater than 86% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RDD strategy (red line).² Ignoring trading costs, a \$1 invested in the RDD strategy would have yielded \$2.57 which ranks the RDD strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RDD strategy would have yielded \$1.54 which ranks the RDD strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RDD relative to those. Panel A shows that

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

the RDD strategy gross alphas fall between the 52 and 67 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RDD strategy has a positive net generalized alpha for five out of the five factor models. In these cases RDD ranks between the 69 and 82 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does RDD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of RDD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RDD or at least to weaken the power RDD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of RDD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDD}RDD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RDD. Stocks are finally grouped into five RDD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RDD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RDD signal in these Fama-MacBeth regressions exceed -0.19, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on RDD is -0.66.

Similarly, Table 5 reports results from spanning tests that regress returns to the RDD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RDD strategy earns alphas that range from 15-19bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.22, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RDD trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.24.

7 Does RDD add relative to the whole zoo?

Finally, we can ask how much adding RDD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the RDD signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes RDD grows to \$1138.20.

8 Conclusion

This study provides robust evidence that Revenue Debt Discipline (RDD) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by [Novy-Marx and Velikov \(2023\)](#), demonstrates that RDD-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of RDD as a return predictor. The value-weighted long/short strategy achieves an annualized

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which RDD is available.

net Sharpe ratio of 0.34, which remains economically significant after accounting for transaction costs. More importantly, the strategy generates statistically significant alphas relative to the Fama-French five-factor model plus momentum, with net returns of 13 basis points per month (t-statistic = 2.01). The robustness of these results is further confirmed by the persistence of abnormal returns even after controlling for six closely related strategies from the factor zoo, including net debt financing, changes in financial liabilities, and various growth measures. The remaining alpha of 15 basis points per month (t-statistic = 2.24) suggests that RDD captures unique information about firm fundamentals not fully explained by existing factors.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The signal’s ability to generate significant net-of-cost returns suggests potential implementability in real-world portfolio strategies, while its orthogonality to existing factors indicates that it may serve as a valuable addition to multi-factor models. The economic intuition behind RDD—that firms exhibiting disciplined revenue-debt relationships are better positioned for future performance—provides a fundamental basis for understanding this anomaly beyond purely statistical patterns.

Several limitations of this study warrant consideration and point to avenues for future research. First, while our analysis follows the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and market conditions examined may not fully capture the signal’s performance across all market regimes. Future research should investigate the time-varying nature of RDD’s predictive power and its performance during different economic cycles. Second, our study focuses primarily on U.S. equity markets; extending the analysis to international markets would provide valuable insights into the global applicability of this signal. Third, while we control for several related factors, understanding the precise economic mechanisms through which RDD affects returns remains an open question worthy of theoretical

development.

Future research directions should also explore the interaction between RDD and corporate financial policies, investigating whether the signal’s predictive power varies across firms with different capital structures or in different industries. Additionally, examining the relationship between RDD and other corporate outcomes, such as default risk, earnings quality, or investment efficiency, could provide deeper insights into the fundamental drivers of this anomaly. Finally, incorporating machine learning techniques to identify non-linear relationships or interaction effects with other firm characteristics may uncover additional dimensions of RDD’s information content.

In conclusion, this paper establishes Revenue Debt Discipline as a robust and economically significant predictor of cross-sectional stock returns, contributing to our understanding of how firm financial discipline relates to expected returns. The signal’s ability to generate significant alphas after controlling for known factors suggests that markets may not fully incorporate information about revenue-debt relationships into prices, presenting both an academic puzzle and a practical opportunity for informed investors.

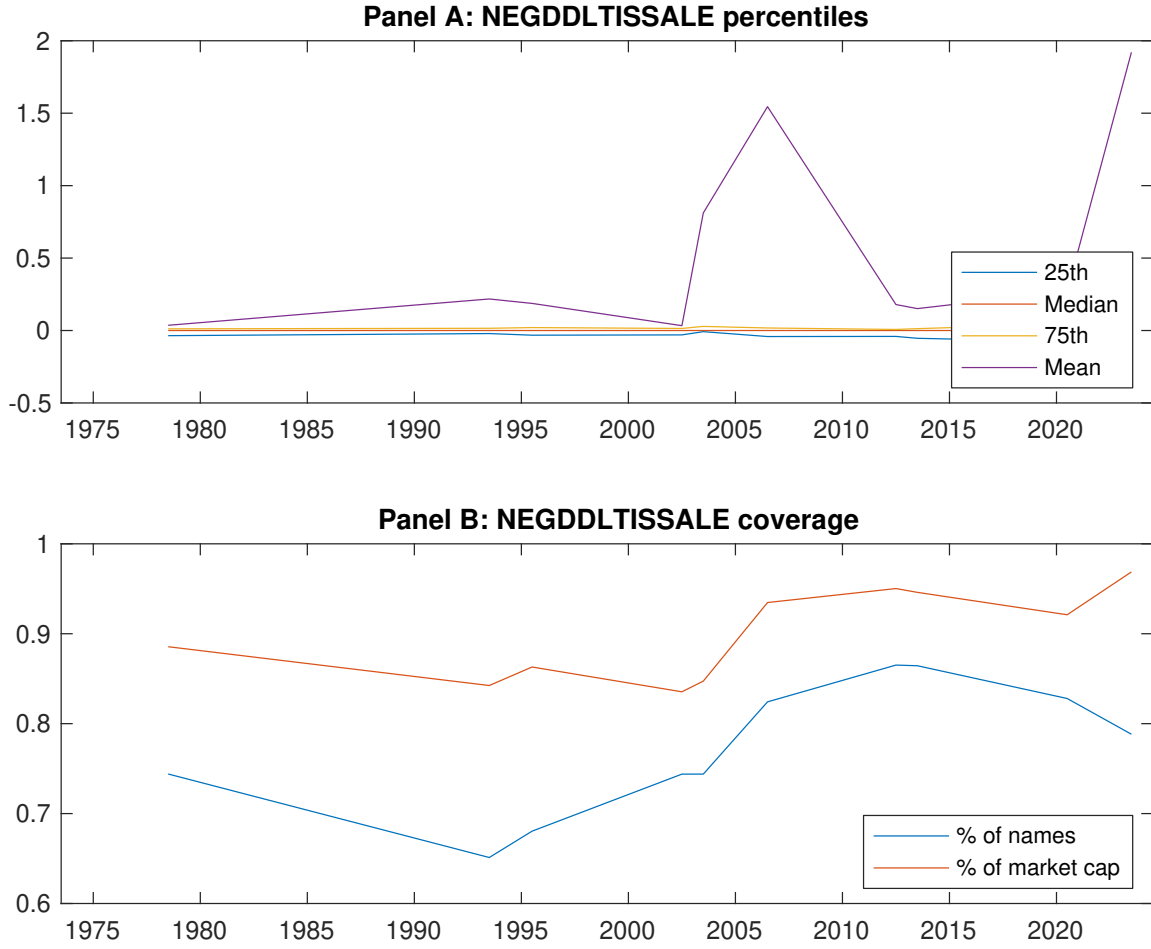


Figure 1: Times series of RDD percentiles and coverage.
This figure plots descriptive statistics for RDD. Panel A shows cross-sectional percentiles of RDD over the sample. Panel B plots the monthly coverage of RDD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RDD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on RDD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.62]	0.66 [3.70]	0.65 [3.25]	0.75 [4.24]	0.77 [3.97]	0.23 [3.27]
α_{CAPM}	-0.16 [-3.09]	0.06 [1.14]	-0.02 [-0.38]	0.15 [3.20]	0.11 [2.16]	0.27 [3.96]
α_{FF3}	-0.20 [-3.83]	0.02 [0.45]	0.05 [0.85]	0.14 [2.92]	0.09 [1.83]	0.29 [4.22]
α_{FF4}	-0.17 [-3.21]	0.05 [0.98]	0.09 [1.57]	0.09 [1.97]	0.08 [1.57]	0.25 [3.58]
α_{FF5}	-0.14 [-2.69]	-0.05 [-1.07]	0.08 [1.46]	0.04 [0.84]	0.06 [1.18]	0.20 [2.96]
α_{FF6}	-0.12 [-2.33]	-0.03 [-0.53]	0.11 [1.95]	0.01 [0.22]	0.06 [1.09]	0.18 [2.62]
Panel B: Fama and French (2018) 6-factor model loadings for RDD-sorted portfolios						
β_{MKT}	1.06 [87.24]	0.97 [89.57]	0.99 [75.37]	0.97 [93.04]	1.02 [84.54]	-0.04 [-2.47]
β_{SMB}	0.02 [0.85]	-0.11 [-6.61]	-0.01 [-0.27]	0.00 [0.27]	0.04 [2.44]	0.03 [1.21]
β_{HML}	0.16 [6.75]	0.10 [4.65]	-0.16 [-6.28]	-0.03 [-1.26]	-0.04 [-1.61]	-0.20 [-6.41]
β_{RMW}	-0.04 [-1.61]	0.16 [7.42]	0.01 [0.26]	0.14 [6.67]	-0.03 [-1.08]	0.01 [0.41]
β_{CMA}	-0.17 [-4.79]	0.07 [2.28]	-0.13 [-3.34]	0.18 [5.81]	0.18 [5.05]	0.35 [7.53]
β_{UMD}	-0.03 [-2.34]	-0.04 [-3.69]	-0.04 [-3.40]	0.04 [4.28]	0.01 [0.55]	0.03 [2.22]
Panel C: Average number of firms (n) and market capitalization (me)						
n	662	526	1031	582	637	
me (\$10 ⁶)	1718	2811	2215	2901	1687	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the RDD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23 [3.27]	0.27 [3.96]	0.29 [4.22]	0.25 [3.58]	0.20 [2.96]	0.18 [2.62]
Quintile	NYSE	EW	0.16 [3.39]	0.18 [3.85]	0.18 [3.78]	0.17 [3.69]	0.16 [3.37]	0.16 [3.41]
Quintile	Name	VW	0.24 [3.48]	0.29 [4.16]	0.31 [4.45]	0.25 [3.66]	0.24 [3.50]	0.21 [3.01]
Quintile	Cap	VW	0.23 [3.37]	0.27 [4.03]	0.29 [4.35]	0.24 [3.57]	0.19 [2.84]	0.16 [2.39]
Decile	NYSE	VW	0.31 [3.35]	0.38 [4.20]	0.38 [4.15]	0.31 [3.43]	0.25 [2.82]	0.22 [2.42]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.17 [2.46]	0.21 [3.06]	0.23 [3.27]	0.20 [2.93]	0.16 [2.27]	0.13 [2.01]
Quintile	NYSE	EW	-0.10 [-1.67]					
Quintile	Name	VW	0.19 [2.68]	0.22 [3.26]	0.24 [3.48]	0.21 [3.08]	0.19 [2.73]	0.16 [2.41]
Quintile	Cap	VW	0.18 [2.61]	0.22 [3.31]	0.24 [3.56]	0.21 [3.16]	0.16 [2.31]	0.13 [2.00]
Decile	NYSE	VW	0.24 [2.58]	0.31 [3.31]	0.30 [3.26]	0.27 [2.89]	0.20 [2.18]	0.17 [1.90]

Table 3: Conditional sort on size and RDD

This table presents results for conditional double sorts on size and RDD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RDD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RDD and short stocks with low RDD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	RDD Quintiles					RDD Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.63 [2.25]	0.98 [3.62]	0.97 [3.63]	1.09 [3.81]	0.70 [2.54]	0.07 [0.91]	0.09 [1.10]	0.08 [0.96]	0.06 [0.80]	0.08 [1.02]	0.08 [0.92]
	(2)	0.74 [2.79]	0.94 [3.73]	0.84 [3.35]	0.93 [3.77]	0.80 [3.19]	0.06 [0.79]	0.10 [1.28]	0.07 [0.86]	0.07 [0.90]	0.04 [0.55]	0.05 [0.62]
	(3)	0.74 [3.00]	0.83 [3.75]	0.85 [3.53]	0.89 [3.90]	0.84 [3.61]	0.10 [1.31]	0.15 [1.91]	0.16 [2.08]	0.13 [1.62]	0.14 [1.83]	0.12 [1.51]
	(4)	0.70 [3.13]	0.82 [3.86]	0.83 [3.70]	0.80 [3.82]	0.84 [3.93]	0.15 [1.92]	0.17 [2.25]	0.17 [2.20]	0.14 [1.82]	0.13 [1.62]	0.11 [1.39]
	(5)	0.42 [2.12]	0.65 [3.62]	0.60 [3.00]	0.70 [3.87]	0.71 [3.78]	0.29 [3.18]	0.33 [3.57]	0.34 [3.69]	0.28 [3.08]	0.21 [2.32]	0.18 [1.99]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	RDD Quintiles					RDD Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	382	384	383	383	379	36	33	31	32	34	
	(2)	106	106	106	106	106	60	60	59	60	59	
	(3)	76	76	76	76	76	107	108	104	106	106	
	(4)	64	64	65	65	64	230	238	229	235	228	
(5)	59	59	59	59	59	1490	2124	1805	2213	1544		

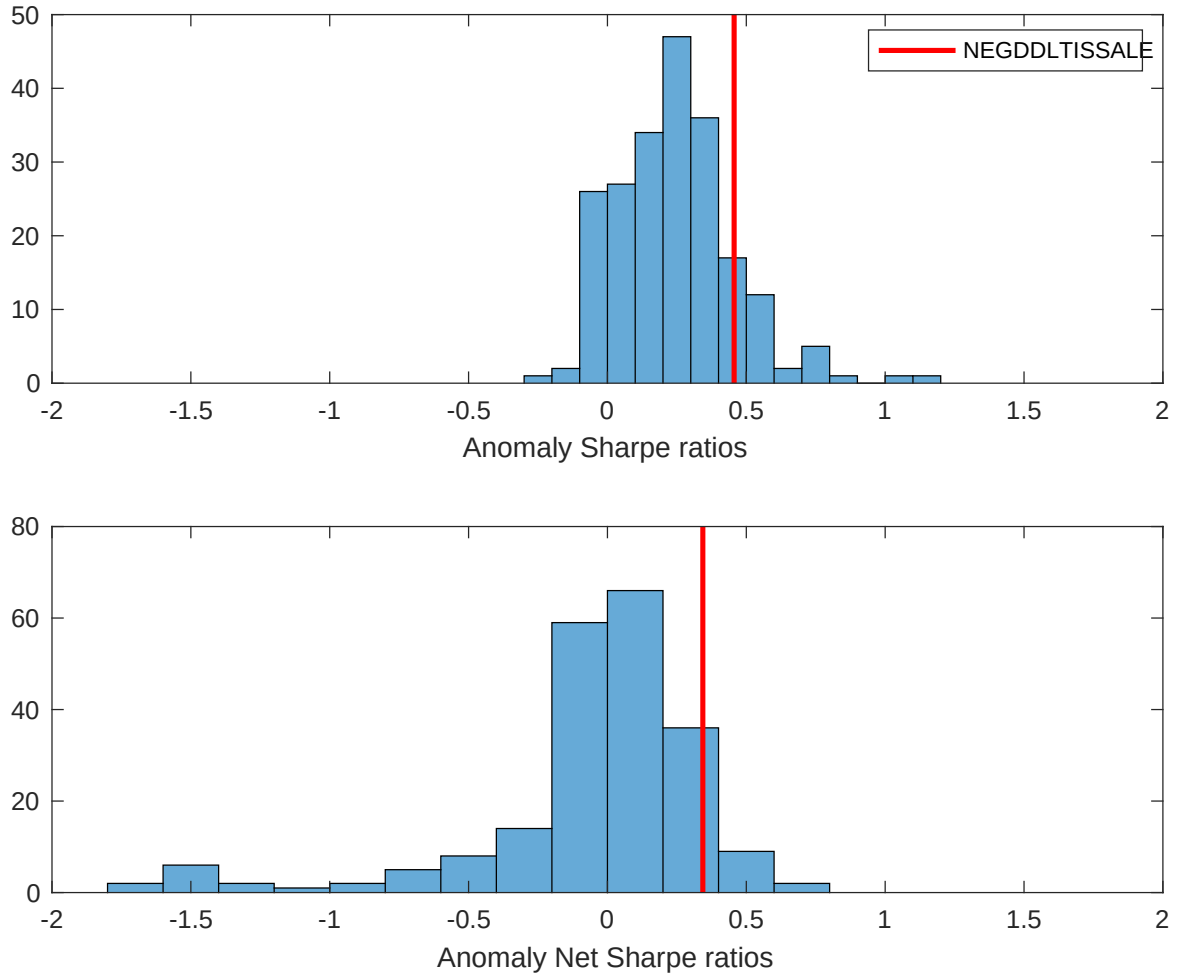


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RDD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

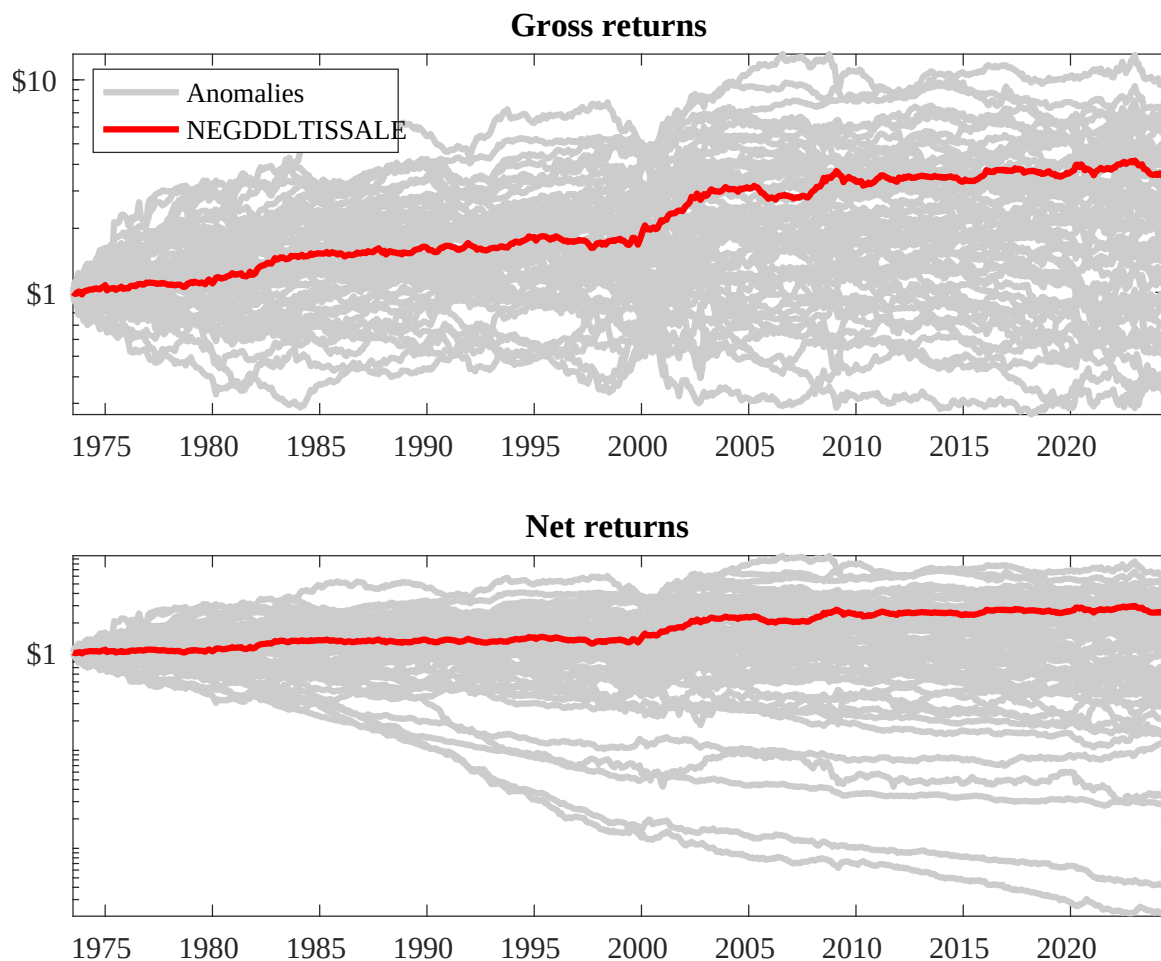
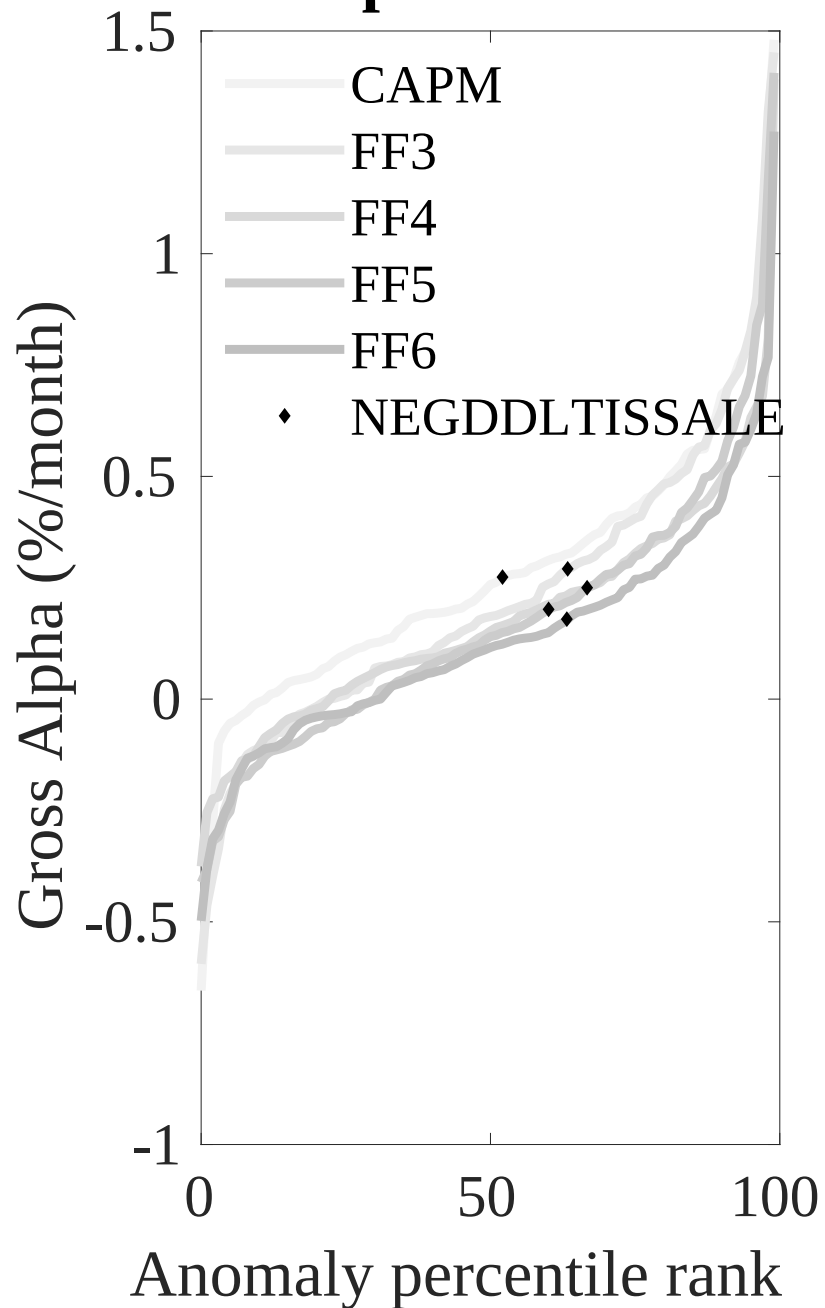


Figure 3: Dollar invested.

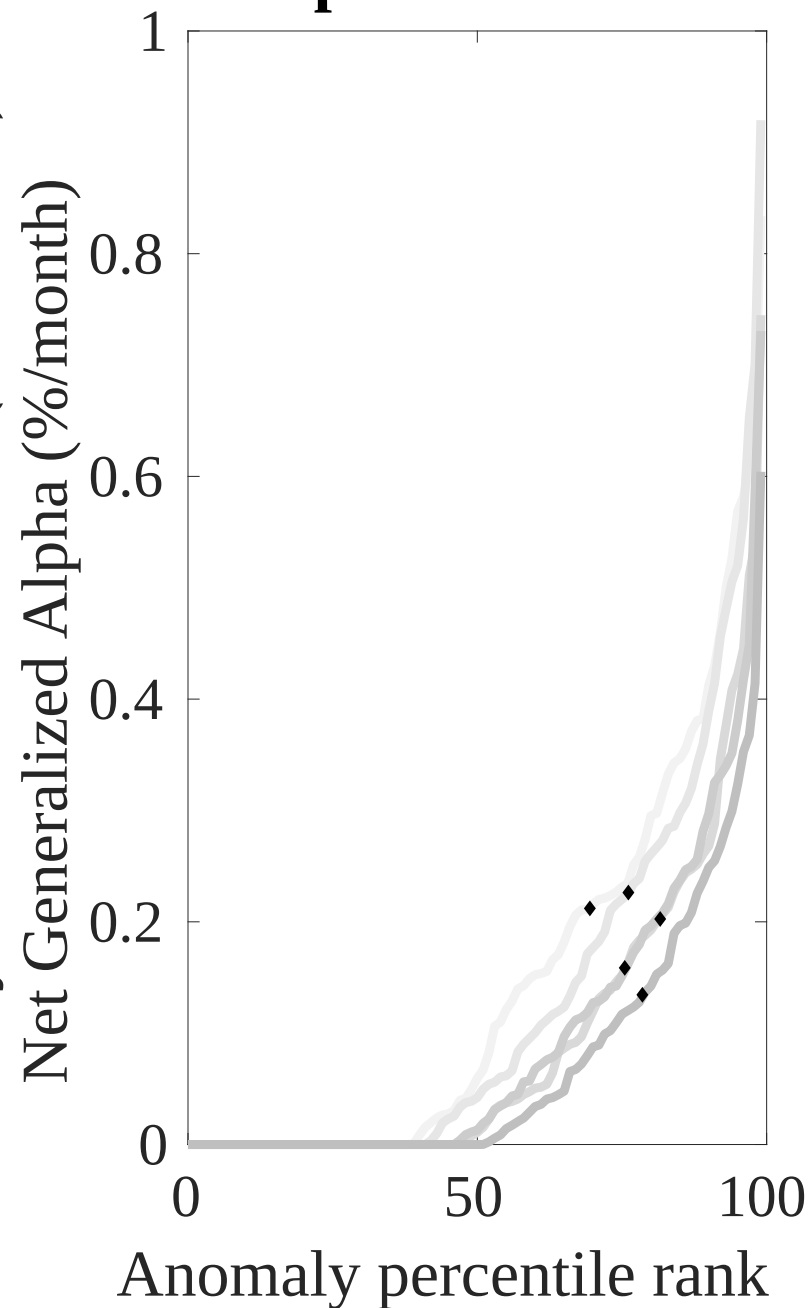
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RDD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



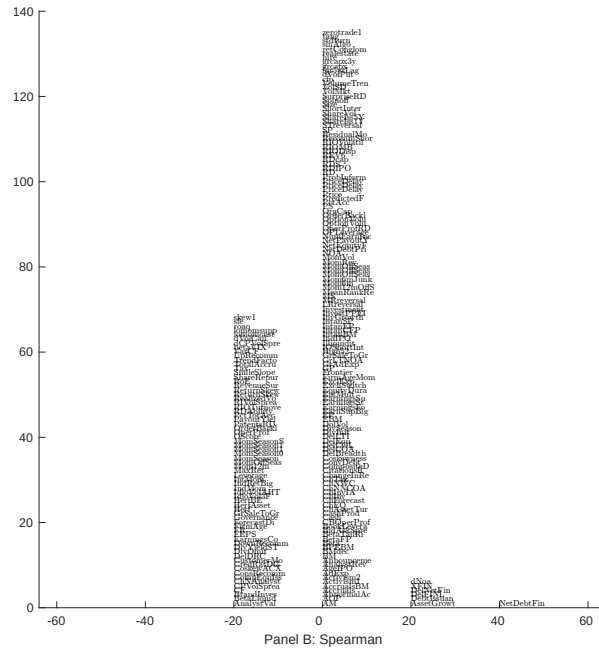
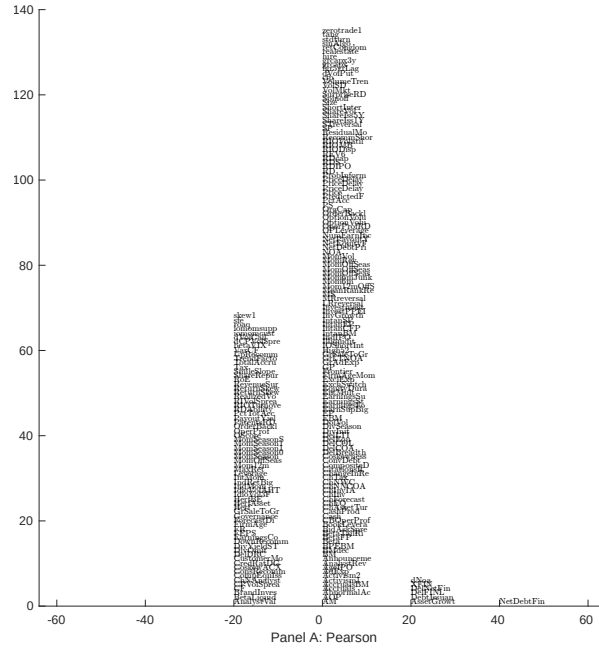


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with RDD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

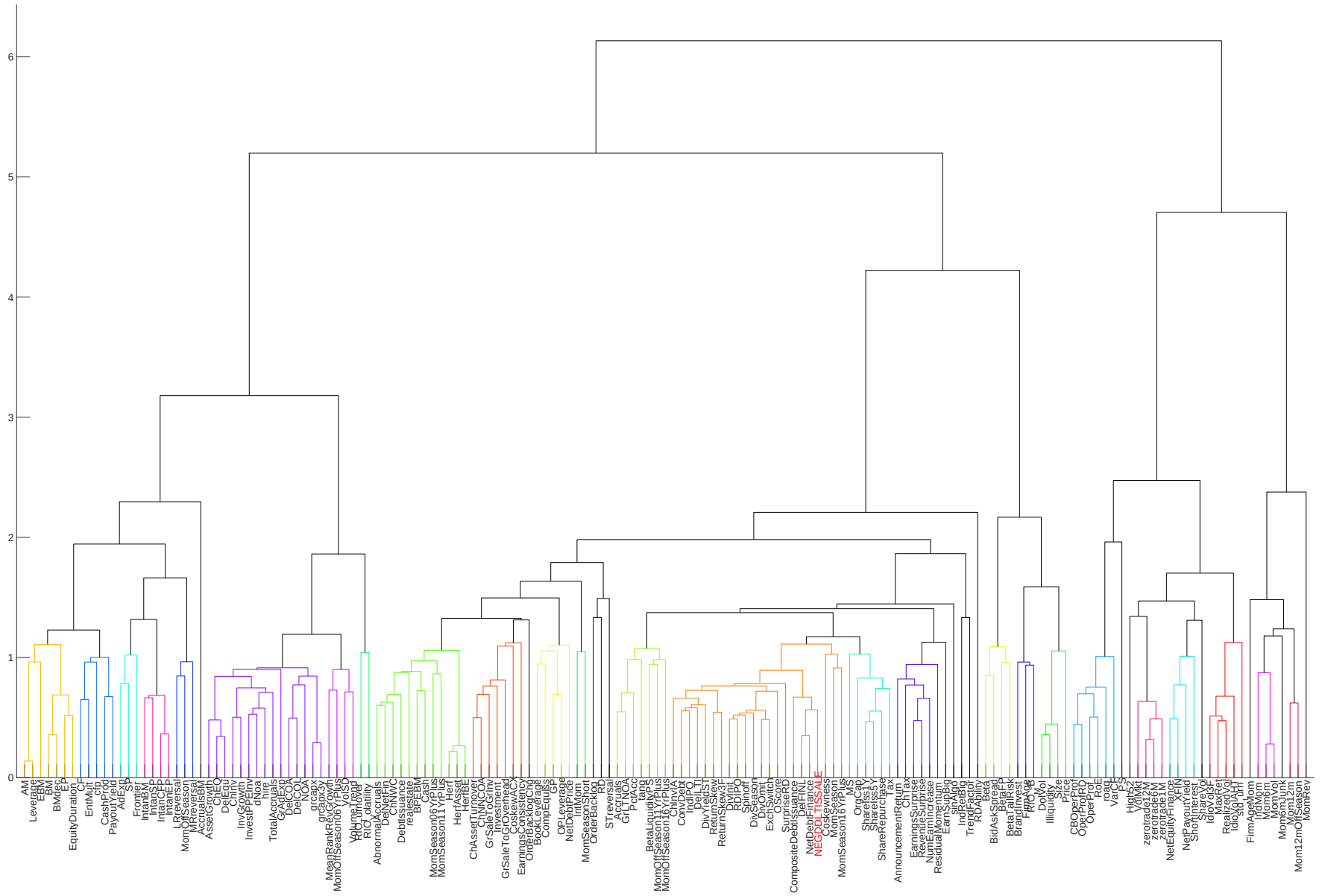


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

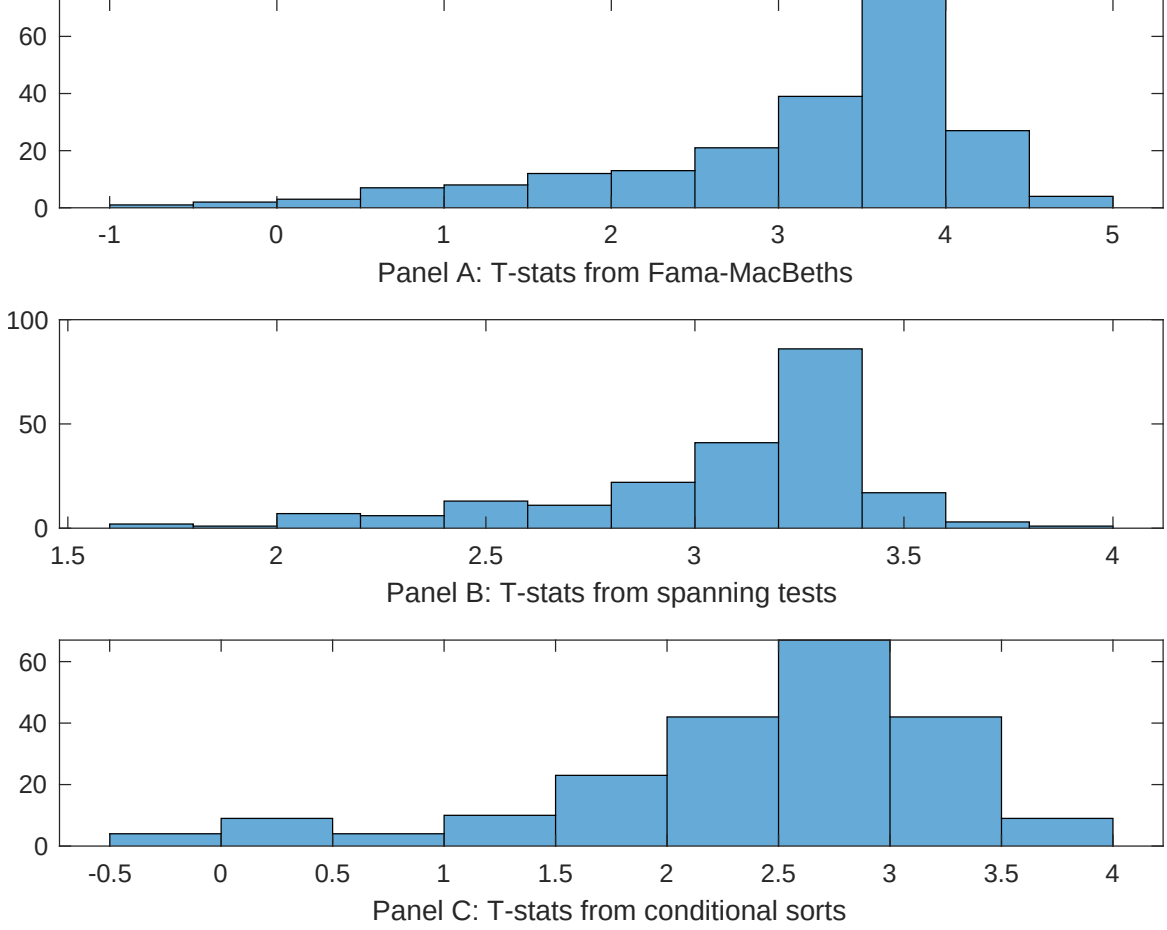


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of RDD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDD}RDD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RDD. Stocks are finally grouped into five RDD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on RDD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RDD}RDD_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.42]	0.13 [5.46]	0.14 [5.70]	0.14 [5.32]	0.14 [5.89]	0.13 [5.42]	0.15 [5.68]
RDD	0.95 [0.14]	0.89 [0.01]	0.68 [1.08]	0.29 [3.55]	-0.13 [-0.19]	0.19 [2.95]	-0.54 [-0.66]
Anomaly 1	0.21 [9.63]						0.96 [1.45]
Anomaly 2		0.18 [9.44]					-0.53 [-1.14]
Anomaly 3			0.20 [6.75]				0.98 [1.68]
Anomaly 4				0.40 [6.87]			0.91 [1.63]
Anomaly 5					0.11 [8.82]		0.76 [3.66]
Anomaly 6						0.94 [6.22]	0.81 [0.58]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the RDD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RDD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.15 [2.28]	0.15 [2.22]	0.15 [2.23]	0.17 [2.54]	0.17 [2.45]	0.19 [2.73]	0.15 [2.24]
Anomaly 1	20.41 [5.60]						11.97 [2.40]
Anomaly 2		17.17 [4.51]					7.28 [1.45]
Anomaly 3			14.69 [4.36]				11.23 [3.12]
Anomaly 4				6.28 [2.37]			4.45 [1.63]
Anomaly 5					5.44 [1.38]		-1.81 [-0.44]
Anomaly 6						6.94 [1.89]	3.61 [0.94]
mkt	-3.56 [-2.33]	-3.21 [-2.07]	-1.59 [-0.98]	-3.72 [-2.38]	-3.61 [-2.30]	-3.42 [-2.18]	-1.83 [-1.13]
smb	1.21 [0.52]	1.25 [0.53]	7.73 [3.00]	3.70 [1.55]	2.62 [1.10]	3.48 [1.46]	5.67 [2.08]
hml	-18.60 [-6.29]	-18.29 [-6.12]	-17.80 [-5.92]	-19.62 [-6.50]	-19.67 [-6.49]	-20.76 [-6.69]	-18.02 [-5.90]
rmw	-0.80 [-0.26]	-0.10 [-0.03]	-7.33 [-2.01]	2.36 [0.76]	1.47 [0.48]	1.76 [0.57]	-6.43 [-1.76]
cma	30.13 [6.57]	29.67 [6.31]	25.85 [5.13]	29.83 [5.80]	28.82 [4.35]	29.23 [5.19]	17.62 [2.56]
umd	2.36 [1.53]	2.19 [1.40]	3.58 [2.33]	3.04 [1.92]	3.80 [2.43]	3.23 [2.05]	1.45 [0.92]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	18	17	16	15	14	14	20

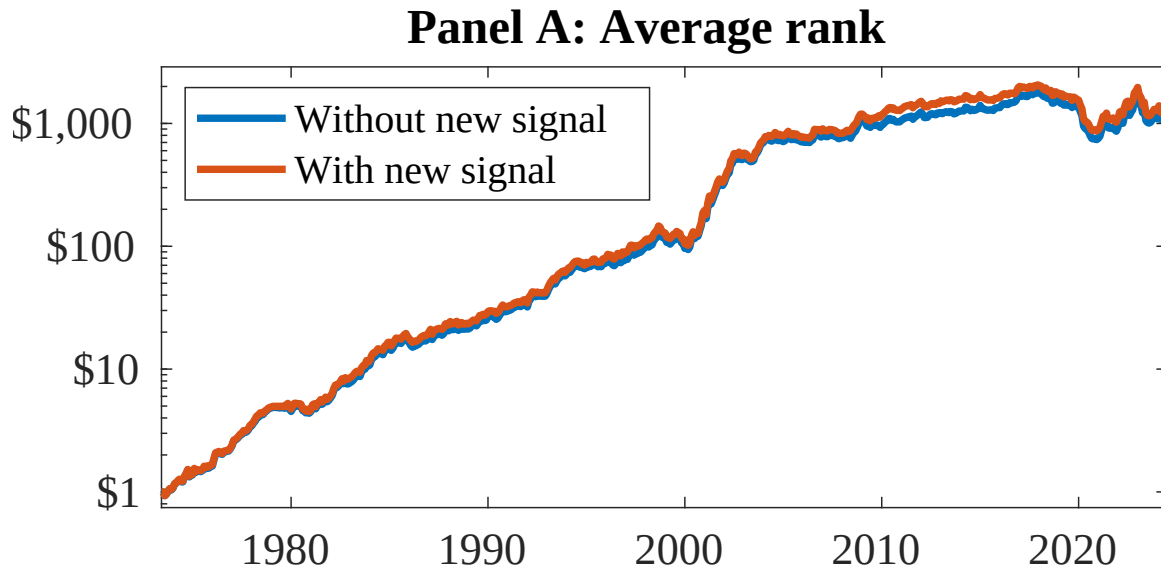


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as RDD. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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